

Day 2:

write A short poem about their neighborhood

Streetlights hum as evening settles in,
Bikes lean against fences, laughter drifting thin,
And every door feels like a story you could walk right in.

wrtie me A professional email asking for a day off

Hi [Manager's Name],

I hope you're doing well. I'd like to request a day off on [date] due to a personal matter. Please let me know if this works with the current schedule, and I'll ensure all my responsibilities are covered.

Thank you for your consideration.

Best regards,

[Your Name]

create a JavaScript function that does something useful

Debounce Function (prevents excessive calls)

This is super helpful for:

- search bars
- API calls
- resize events
- typing input

```
function debounce(fn, delay = 300) {  
  let timeoutId;
```

```
return function (...args) {  
  clearTimeout(timeoutId);  
  
  timeoutId = setTimeout(() => {  
    fn.apply(this, args);  
  }, delay);  
};  
}
```

What it does

It makes sure a function only runs **after the user stops triggering it**.

```
const searchInput = document.getElementById("search");
```

```
function handleSearch(event) {  
  console.log("Searching for:", event.target.value);  
}
```

```
const debouncedSearch = debounce(handleSearch, 500);
```

```
searchInput.addEventListener("input", debouncedSearch);
```

💡 Example (search input)

```
const searchInput = document.getElementById("search");
```

```
function handleSearch(event) {
```

```
    console.log("Searching for:", event.target.value);  
  }  
  
const debouncedSearch = debounce(handleSearch, 500);  
  
searchInput.addEventListener("input", debouncedSearch);
```

The Three Elements of a Good Prompt

1. Context
2. Specificity: In recording file
2. Iteration: In recording file

Why Prompts Matter

Create a simple HTML page with:

- A centered heading that says "Welcome to My Portfolio"
- A navy blue background (#1a1a2e) with white text
- Three sections: About Me, My Skills, Contact
- Each section should have a heading and a short paragraph of placeholder text
- Use a clean, modern font (Arial or Helvetica)

1. See Why_Prompts_Matter recording file.

Lab Exercise 1: Prompt Makeover

I was assigned to Bad Prompt B in my group: I asked Claude with the bad prompt: Help me with my Resume. Claude went on to ask me a series of question and provide me with selections. It asked is this a rewrite or starting from scratch? Then asked is this for front-end, full-stack, software-engineer. It asked priorities, I selected all of the above and I provided it with my file because I chose re-write.

Claude was surprisingly good at revising my resume. It relabeled my "about me" section and added technical skills. Then it relabeled my objective and changed into "summary". Then, it tailored my resume to the chosen selections from the drop down it gave me.

I then used up my last tokens with ChatGPT and ran the improved prompt. I prompted ChatGPT to revise my resume and use the previous bio it created for me from the previous exercise. I chose this bio because it was impressive. I told ChatGPT to only improve the sections that needed to emphasize how my insurance background and behavioral detection and analysis background was relative to software engineering. Chat GPT only returned three sections. The objective (which was changed to summary), and the employment sections which were revised with the bio content from the previous exercise. It was an impressive revision. I would gladly take the information and replace those sections from Chat GPT but keep the layout that Claude gave me with the bad prompt.

Lab Exercise 2: Reading Code

Write a simple JavaScript program that:

- Asks the user for their name
- Asks the user for their favorite color
- Displays a greeting message that includes both

Use `prompt()` for input and `document.write()` for output.

Add comments explaining each line.

Since I ran out of tokens, I am using Claude. Claude created a window prompt asking me what is my name? The html doesn't even include a placeholder in HTML code because AI used JavaScript to create the prompt under the variable constant name and color. When I answered the question, the next variable `const color` asked me What is my favorite color? Then, when I looked at the preview It used `document.write` with an HTML `<p>Hello, (name) !`" and then the variable `(name)`, Your favorite color is `(color).``<p>` for paragraph, inserted in javascript, displaying, Hello, Jacqueline! Your favorite color is green.

If I typed Marcus and green. It says Hello, Marcus! Your favorite color is green.

If I clicked cancel, it would say Hello, null! Your favorite color is null.

When I tried to modify the code, the prompt kept appearing and it consistently kept interfering as I tried to update the code. Yes it worked because AI was asked to use `prompt()` which returns a prompt or a window prompt. Under

document.write it must include the dollar sign `$()` before the parentheses for the variable.

```
<!DOCTYPE html>

<html>

<body>

<script>

  // Ask the user for their name and store the response
  const name = prompt("What is your name?");

  // Ask the user for their favorite color and store the response
  const color = prompt("What is your favorite color?");

  // Display a greeting that includes both pieces of information
  document.write(`<p>Hello, ${name}! Your favorite color is ${color}.</p>`);

</script>

</body>

</html>
```

Peer Activity: Prompt Partners

Michael's worked better. We discovered that using Gemini pro was yielding better results than Chat GPT. Michael used similar prompts than I did for the dog walking business but he used words like minimalist and professional in his prompt which returned a better result. Even when you used less information, Gemini was Pro and fast were better at output. Chat GPT definitely needed more information to receive a better result.

Code Reading + Modification Lab: Exercise 1

Claude didn't display the javascript with the prompt "Write a javascript tip calculator that asks for the bill amount and tip percentage, then shows the tip and total." Claude displayed the calculator seen below on the next page.

```
let tipPct = 18;
```

```
function fmt(n) {  
    return '$' + n.toFixed(2);  
}
```

```
function calc() {  
    const bill = parseFloat(document.getElementById('bill').value) || 0;  
    const people = parseInt(document.getElementById('split').value) || 1;  
    const tip = bill * tipPct / 100;  
    const total = bill + tip;  
    document.getElementById('tip-out').textContent = fmt(tip);  
    document.getElementById('total-out').textContent = fmt(total);  
    document.getElementById('per-person').textContent = fmt(total / people);  
}
```

```
function setTip(val) {  
    tipPct = val;  
    document.getElementById('tip-slider').value = val;  
    document.getElementById('pct-label').textContent = val + '%';  
    highlightPreset(val);  
    calc();  
}
```

```
function highlightPreset(val) {  
  document.querySelectorAll('#presets button').forEach(btn => {  
    const active = parseInt(btn.dataset.val) === val;  
    btn.style.background = active ? 'var(--color-background-info)' : 'transparent';  
    btn.style.color = active ? 'var(--color-text-info)' : 'var(--color-text-primary)';  
    btn.style.borderColor = active ? 'var(--color-border-info)' : 'var(--color-border-secondary)';  
  });  
}
```

```
document.getElementById('tip-slider').addEventListener('input', function() {  
  tipPct = parseInt(this.value);  
  document.getElementById('pct-label').textContent = tipPct + '%';  
  highlightPreset(tipPct);  
  calc();  
});
```

```
document.getElementById('bill').addEventListener('input', calc);  
document.getElementById('split').addEventListener('input', calc);
```

```
highlightPreset(18);
```

Bill amount

\$ 0.00

Tip percentage — 18%

10% 15% 18% 20% 25%

Tip \$0.00

Total \$0.00

Split between 1 people

Each person pays \$0.00

But I took the code from Exercise 1: Tip Calculator and inserted it in Replit. See the recording for the js code prompt calculator. The challenges change it split between any number of people: `parseInt(prompt("How many people are splitting the bill?"))`; and Level 3 because I wasn't sure where to enter the conditional. Now that I chatted with AI, I learned where it should be placed which makes sense because you wanted it to be under `document.write` but I couldn't get past the `{ }` brackets. I wanted to enter `if tip > 20` `document.write("<p> ...)` but I had to use `{ }`

```
if (tip > 20) { document.write("<p>Generous tipper!</p>"); }
```

Exercise 2: Temperature Converter

Ask Chat GPT "Write a JavaScript program that converts Fahrenheit to Celsius. Use the formula: $C = (F - 32) \times 5/9$. Ask the user for the Fahrenheit temperature and display the Celsius result."

Claude:


```
<!DOCTYPE html>

<html>

<body>

<script>

    // Ask the user for the Fahrenheit temperature
    const fahrenheit = parseFloat(prompt("Enter temperature in Fahrenheit:"));

    // Convert to Celsius using the formula  $C = (F - 32) \times 5/9$ 
    const celsius = (fahrenheit - 32) * 5 / 9;

    // Display the result
    document.write("<h2>Temperature Converter</h2>");
    document.write("<p>" + fahrenheit + "°F = " + celsius.toFixed(1) + "°C</p>");
</script>

</body>

</html>
```